LABORATORY WORK NO. 2

Cryptanalysis of Monoalphabetic Ciphers

Breaking an intercepted cryptogram by frequency analysis

Discipline: **Cryptography**

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Variant: **16**

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1. Short theory

A monoalphabetic cipher replaces every occurrence of a letter with the same substitute throughout the message. Its key is an arbitrary permutation of the alphabet, so the key space is $26! \approx 4.03 \times 10^{26}$ — far too large to search. The cipher is nevertheless broken routinely, because the key space is not where its weakness lies.

The weakness is that the substitution leaves the *structure* of the language untouched. Each plaintext letter maps to one ciphertext letter and always the same one, so everything that makes a letter recognisable survives into the ciphertext: how often it occurs, which letters it sits next to, which short words it forms, whether it doubles. The attacker does not need the key — the text carries enough of the language to give the key away.

Signal	What it gives away
Letter frequencies	`E` is the most common letter of English at 12.70%, then `T`, `A`, `O`, `I`, `N`
Short words	The commonest three-letter word is `THE`; two-letter words are mostly `OF`, `TO`, `IN`, `IS`, `AS`
Bigrams	`TH`, `HE`, `AN`, `IN`, `ER`, `ON`, `RE`, `ED`, `ND`, `HA`, `AT`, `EN`
Trigrams	`THE`, `AND`, `THA`, `ENT`, `ION`, `TIO`, `FOR`, `NDE`
Doubled letters	`SS`, `EE`, `TT`, `OO` and `FF` are the frequent ones

The attack works by hypothesis and confirmation rather than by calculation. The frequency order of the cryptogram is lined up against that of the language, which suggests a few substitutions; those are made, and the partly decrypted text is read to see which words are beginning to appear. A word that becomes readable fixes several more letters at once, and each new letter makes more words readable. The process converges quickly and is self-checking — a wrong guess produces unpronounceable text and is abandoned.

Two conditions have to hold: the message must be long enough for its letter frequencies to resemble those of the language, and the language of the plaintext must be known. Both are satisfied here.

2. The intercepted cryptogram

Variant 16. The message is 3985 characters long, of which 3239 are letters. Two things are known about it: it was enciphered with a monoalphabetic cipher, and the plaintext is English. Only the letters were enciphered — digits, punctuation, spaces and the capitalisation are exactly as they were in the original, which is a considerable help, because the word boundaries are visible.

```
Wqv cxipw ngv rtp rtxwxgj xg wqv rxgjp. Wqxp rtp unsf –tsuqtavwxhpdapwxwdwxng, xg wqv cniz nc wqv
pwitxjqw–tsuqtavw Kxjvgviv rxwq pqniwivuvtwxgj lvfrnio. Wqv nso naevhwxngp wn xwp dpv, rqxhq anxsvo
onrg
wnwqv xzunppxaxsxwf nc hniivhwxgj t jtiasvo oxputwhq bdxhlsf vgndjq,ktgxpqvo rxwq wqv wvsvjituq. Xw
cdscxssvo wqv ivbdxivzvgwp nc gnghnz –uinzxptaxsxwf nc wq v jvgvits pfpwvz tgo nc vtpv nc lvf hqtgjvp.
Znivnkvi,xw qto wqv ivudwtwxng nc avxgj dgaivtltasv–rqxhq, xc xwp hifuwnjitzprviv gnw oxkxovo xgwn
rniop, xw stijvsf rtp. Wqv zxsxwtif tonuwvo xw twnghv.Wqvg, xg 1863, t ivwxivo Uidppxtg xgctgwif zteni
oxphnkviv o wqvjvgvits pnsdwxng cni wqv uvixnoxh unsftsuqt –avwxh pdapwxwdwxng. Tw ngvpwinlv qv
ovznsxpqvo wqv ngsf xzuivjgtasv pwidhwdiv xg hifuwnjituqf.Pxjgts nccxhvip, hnzuvssvo wn uinkxov pvhdiv
hnzzdgxhtwxngp, qdgwvocitgwxhtssf cni gvr cxvso hxuqvip. Wqvf cndgo ztgf jnno xovtp xg wqvrixwxgjp nc
wqv oxsvwwtgwv hifuwnjituqvip rqn qto uinunpvo hxuqvip cniwqv uinwvhwxng nc uixktwv zvpptjvp. Pnng pnzv
nc wqvpv pfpwvzp rvivpvikxgj xg wqv ktixndp tizxvp nc Vdinuv tgo wqv Tzvixhtp. Zniv xovtphtzv cinz tizf
nccxhvip rqn qto pwdoxvo hifuwnjituqf xg wqv hndipvp xgpxjgts hnzzdgxhtwxng wqtw wqv gtwxngts zxsxwtif
thtovzxvp, pdhq tp Pw.Hfi, qto toovo xg wqv zxo–1800p. Xgvkxwtasf, hifuwtgtsfpwp–rqn rvivvxwqvi
tztwvdip
```

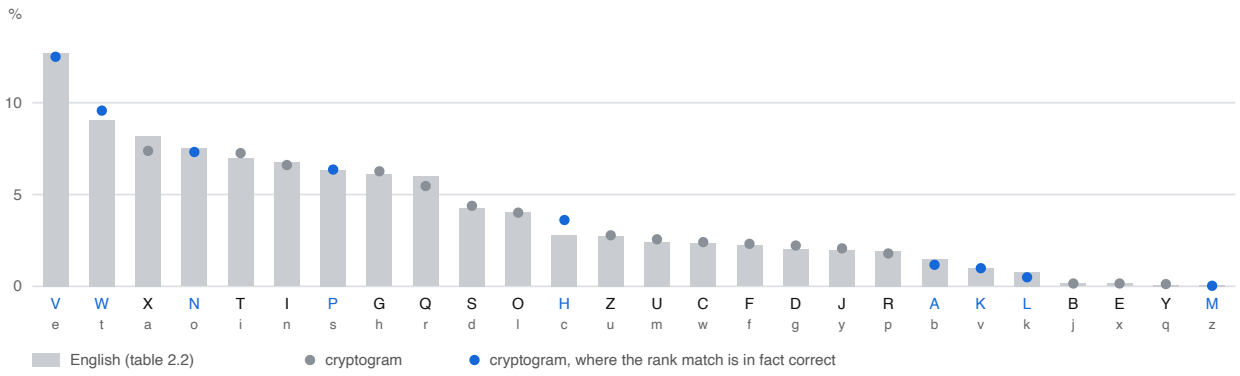
```
ni pnsoxvip rxwq t uincvppxngts xgwvivpw, cni cdssuincvppxngtsp wqviv rviv gngv—ivusxvo rxwq gvr
wvhqgxbdvp cni aivtlxgjwqv gvr hxuqvip. Cinz wqv psnr hitrs nc gnzvghstwni otfp, rqvg wqvxgwinodhwxng
nc
t puvhxts jindu zvtgxgj Oxpivjtio wqv uivhvoxgj jindurndso hngpwxwdwv t ivztiltasv wvhqgxhts toktghv,
wqv ithv avwrvvgnccvgpv tgo ovcvgpv xg hifuwnsnjf thhsvitwvo wn xwp znovig uthv.Wqv qxpwnif nc
hifuwnsnjf cinz wqv ovhtov wqtw ptr anwq wqv ovtwq ncwqv asthl hqtzavip tgo wqv axiwq nc wqv wvsvjituq
wn Rniso Rti X xp wqdpt pwnif nc xgwvigts ovkvsnuzvgw. Rxwqndw Inppxjgns p ni Rxssvpvp, tgorxwqndw
zteni
rtip ni oxusnztwxh pwidjjsvp, hifuwnsnjf hndso gnwxgcsdvghv rniso vkvgwp, tgo, vyhvuw cni ngv ni wrn
dgdpdts htpvp, xw oxognw. Wqv wvsvjituq stdghqvo wqxp vknsdwxng nc hifuwnsnjf. Xw ainlv wqv zngnunsf nc
wqv gnzvghstwni. Wq vgnzvghstwni qto ivxjgvo cni 450 fvtip tp t jvgvits, tss—udiunpv pfpwvz,adw xw
hndso
gnw zvvw wqv gvr ivbdxivzvgwp vxwqvi nc qxjq —svkvsoxusnztwxh ni zxsxwtif hnzzdgxhtwxngp ni nc snr
—svkvs pxjgtshnzzdgxhtwxngp, rqxhq wqv wvsvjituq qto vgjvgovivo. Vthq htss vo cni xwpnrg lxgo nc
hifuwnpfpwvz, t puvhxtsxmvo ngv. Pxjgts nccxhvip itglvo wqvpvpfpwvzp xg t qxvitihqf, ixpxgj cinz wqv
pxzusv tgo csvyxasv tgo vtpxsfpnskvo wn wqv vywvgpxkv tgo qtio wn pnskv. Wqv wvsvjituq wqdp
pwxzdstwvowqv xgkvgwxng nc ztgf gvr hxuqvip tgo, af ivthwxng, ztgf gvr zvwqnopnc hifuw— tgtsfpxp, tgo
hnzuvssvo wqvxi tiitgjvzvgw xg t phtsv nchnzusvyxwf.Ztgf nc wqvpv hxuqvip tgo wvhqgxbdvp qtkv avhnzv
hstppxh tgo tiv xgdpv wnotf. Znivnkvi, hifuwnjituqf pwxss cdghwxngp wqindjq t qxvitihqftgo vzusnfp t
zdswxwdov nc puvhxts pfpwvzp. Wqv wvsvjituq wqvivafcdigxpqvo hifuwnjituqf rxwq wqv pwidhwdiv tgo wqv
hngwvgw wqtw xw pwxssqtp. Xw ztov xw rqtw xw xp wnotf. Tss wqvpv wqxgjp qtkv tgwvhvovgwp, tgo edpw tp
wqv wvsvjituq xwpvsc oxo,pn rviv wqviv uivhdipnip nc wqv hifuwnjituqxh pfpwvzp wqtw xwvgjvgovivo.Ngv
hxuqvi pfpwvz xgkvgwvo avcniv wqv wvsvjituq rtp pn cti tqvto ncxwp wxzv, tgo pn zdhq xg wqv puxixw nc
wqv stwvi xgkvgwxngp, wqtw xw ovpvikvpwn av hstppvo rxwq wqvz. Xgovvo, xw ovpvikvp wqv cingw itgl tzngj
wqvz,cni wqxp pfpwvz rtp avfngo ondaw wqv znpw ivztiltasv nc tss. Pn rvsshnghvxkvo rtp xw wqtw wnotf,
zniv wqtg t hvgwdif tgo t qtsc nc ituxowvhqgnsnjxhts uinjivpp tcwvi xwp xgkvgwxng, xw ivztxgp xg thwxkv
dpv.Adw wqvg xw rtp xgkvgwvo af t ivztiltasv z tg, t rvss —lgnrg rixwvi,tjixhdswditsxpw, axasxnuqxsv,
tihqxwvhw, oxusnztw, jtojvwvvi, tgo pwtwvpztggtzvo Wqnztp Evccvipng. Qv htssvo xw qxp "rqvvs hfuqvi,"
tgo xw pvvzpsxlvsf wqtw qv xgkvgwvo xw vxwqvi odixgj 1790 wn 1793 ni odixgj 1797 wn1800.
```

A few words are run together (`wqvjvgvits`, `wqvrixwxgjp`) and a few have a stray space inside them (`wq v`, `htss vo`). These come from the way the text was laid out in the assignment, not from the cipher, and they do not interfere with the attack.

3. Frequency analysis

Counting the letters of the cryptogram and placing them in decreasing order gives a profile that follows English very closely in shape. `V` leads at 12.50% against 12.70% for `E`, and the curve falls away at the same rate.

Rank	Cipher	%	English	%	Rank	Cipher	%	English	%
1	V	12.50	E	12.70	14	U	2.56	M	2.41
2	W	9.57	T	9.06	15	C	2.41	W	2.36
3	X	7.38	A	8.17	16	F	2.32	F	2.23
4	N	7.32	O	7.51	17	D	2.22	G	2.01
5	T	7.26	I	6.97	18	J	2.07	Y	1.97
6	I	6.61	N	6.75	19	R	1.79	P	1.93
7	P	6.36	S	6.33	20	A	1.17	B	1.49
8	G	6.27	H	6.09	21	K	0.99	V	0.98
9	Q	5.46	R	5.99	22	L	0.49	K	0.77
10	S	4.38	D	4.25	23	B	0.15	J	0.15
11	O	4.01	L	4.03	24	E	0.15	X	0.15
12	H	3.61	C	2.78	25	Y	0.12	Q	0.09
13	Z	2.78	U	2.76	26	M	0.03	Z	0.07



Letter frequencies of the cryptogram against English, by rank. Bars: English (table 2.2). Dots: the cryptogram. Blue marks the ranks at which matching the two orders happens to give the correct letter — 9 of 26.

The chart shows why counting alone does not finish the job. Lining the two orders up rank for rank is correct for only 9 of the 26 letters — but it is correct for the first four, and those four are the ones worth having. `E`, `T`, `A` and `O` account for roughly 38% of English text, so fixing them makes a large part of the message readable, and the rest of the attack is driven by the words that then appear rather than by further counting.

The other signals

Word boundaries survived the encipherment, so short words and repeated groups are visible too:

Signal	Most frequent in the cryptogram
Three-letter words	`WQV` (43), `TGO` (18), `CNI` (8), `RTP` (7), `QTO` (6), `GVR` (6)
Two-letter words	`NC` (28), `XW` (19), `XG` (13), `WN` (8), `NI` (7), `PN` (4)
One-letter words	`T` (15), `P` (2)
Trigrams	`WQV` (67), `XNG` (20), `TGO` (20), `WXN` (18), `JIT` (15), `ITU` (15)
Doubled letters	`SS` (15), `PP` (9), `VV` (7), `CC` (5), `ZZ` (4), `II` (2)

Two entries stand out before any substitution is made. `WQV` occurs 43 times as a word and 67 times as a trigram — no English word other than `THE` behaves like that. And `T` standing alone 15 times can only be `A` or `I`, since those are the only one-letter words in English.

4. Breaking the cipher

The attack proceeds exactly as in section 2.3 of the assignment. After every deduction the substitutions found so far are applied and the result is read back. Letters already recovered appear in **lowercase**; letters still unknown stay as the **UPPERCASE** ciphertext letter, so a half-broken word such as `tIe` shows at a glance what is missing.

The running key is written as 26 characters — the plaintext letter that each ciphertext letter `a` to `z` stands for, with a dot where nothing has been decided yet. Only the opening of the message is reproduced at each step; `python3 main.py walkthrough` prints all eight stages in full.

Step 1 — the giveaway word

The three-letter group WQV occurs 43 times as a word and 67 times as a trigram - far more than anything else in the cryptogram. The commonest three-letter word in English is THE, and V is also the most frequent letter here (12.50%, against 12.70% for E in English).

```
key: ...............h....et...

the CXIPt NGe RTP RTXtXGJ XG the RXGJP. thXP RTP UNSF –TSUhTAetXHPDAPtXtDtXNG, XG the CNIZ NC the
PtITXJht–TSUhTAet KXJeGeIe RXth PhNItIeUeTtXGJ LeFRNIO. the NSO NAEeHtXNGP tN XtP DPe, RhXHh ANXSeO
ONRG
tNthe XZUNPPXAXSXtF NC HNIIeHtXGJ T JTIASeO OXPUTtHh BDXHLSF eGNDJh,KTGXPheO RXth the teSeJITUh. Xt
CDSCXSSeO the IeBDXIeZeGtP NC GNGHNZ –UINZXPTAXSXtF NC th e JeGeITS PFPteZ TGO NC eTPe NC LeF HhTGJeP.
```

Step 2 — TO and OF

WN occurs 8 times and now reads T-. The only common English two-letter word beginning with T is TO, so N is O - which agrees with N being the fourth most frequent letter at 7.32%, against 7.51% for O. NC, the most frequent two-letter word at 28 occurrences, then reads O- and must be OF.

```
key: ..f.........o..h....et...

the fXIPt oGe RTP RTXtXGJ XG the RXGJP. thXP RTP UoSF –TSUhTAetXHPDAPtXtDtXoG, XG the foIZ of the
PtITXJht–TSUhTAet KXJeGeIe RXth PhoItIeUeTtXGJ LeFRoIO. the oSO oAEeHtXoGP to XtP DPe, RhXHh AoXSeO
OoRG
tothe XZUoPPXAXSXtF of HoIIeHtXGJ T JTIASeO OXPUTtHh BDXHLSF eGoDJh,KTGXPheO RXth the teSeJITUh. Xt
fDSfXSSeO the IeBDXIeZeGtP of GoGHoZ –UIoZXPTAXSXtF of th e JeGeITS PFPteZ TGO of eTPe of LeF HhTGJeP.
```

Step 3 — A and AND

T stands alone as a word 15 times, and English has only two one-letter words, A and I. TGO is the second most frequent three-letter word (18 times), exactly as AND is in English; reading T as A makes TGO fit AND and fixes two more letters.

```
key: ..f...n......od.h..a.et...

the fXIPt one RaP RaXtXnJ Xn the RXnJP. thXP RaP UoSF –aSUhaAetXHPDAPtXtDtXon, Xn the foIZ of the
PtIaXJht–aSUhaAet KXJeneIe RXth PhoItIeUeatXnJ LeFRoId. the oSd oAEeHtXonP to XtP DPe, RhXHh AoXSed
doRn
tothe XZUoPPXAXSXtF of HoIIeHtXnJ a JaIASed dXPUatHh BDXHLSF enoDJh,KanXPhed RXth the teSeJIaUh. Xt
fDSfXSSed the IeBDXIeZentP of nonHoZ –UIoZXPaAXSXtF of th e JeneIaS PFPteZ and of eaPe of LeF HhanJeP.
```

Step 4 — IN and IT

XG (13 times) now reads -N. It cannot be AN, because A is already T, so X is I and XG is IN. This is confirmed at once by XW, the second most frequent two-letter word at 19 occurrences, which now reads IT.

```
key: ..f...n......od.h..a.eti..

the fiIPt one RaP RaitinJ in the RinJP. thiP RaP UoSF –aSUhaAetiHPDAPtitDtion, in the foIZ of the
PtIaiJht–aSUhaAet KiJeneIe Rith PhoItIeUeatinJ LeFRoId. the oSd oAEeHtionP to itP DPe, RhiHh AoiSed
doRn
tothe iZUoPPiAiSitF of HoIIeHtinJ a JaIASed diPUatHh BDiHLSF enoDJh,KaniPhed Rith the teSeJIaUh. it
fDSfiSSed the IeBDiIeZentP of nonHoZ –UIoZiPaAiSitF of th e JeneIaS PFPteZ and of eaPe of LeF HhanJeP.
```

Step 5 — TELEGRAPH

WVSVJITUQ appears 8 times and reads TE-E--A-H. The repeated E in that pattern and the subject of the passage leave only one reading: TELEGRAPH.

```
key: ..f...n.rg...od.h.lapeti..

the firPt one RaP Raiting in the RingP. thiP RaP polF –alphaAetiHPDAPtitDtion, in the forZ of the
Ptraight–alphaAet Kigenere Rith Phortrepeating LeFRord. the old oAEeHtionP to itP DPe, RhiHh Aoiled
doRn
tothe iZpoPPiAilitF of HorreHting a garAled diPpatHh BDiHLlF enoDgh,KaniPhed Rith the telegraph. it
fDlfilled the reBDireZentP of nonHoZ –proZiPaAilitF of th e general PFPteZ and of eaPe of LeF HhangeP.
```

Step 6 — WAS, AS and IS

RTP occurs 7 times and reads -AS, which can only be WAS. That makes P equal to S, and the two-letter words TP and XP then read AS and IS, both of which are common English words.

```
key: ..f...n.rg...odshwlapeti..

the first one was waiting in the wings. this was polF –alphaAetiHsDAstitDtion, in the forZ of the
straight–alphaAet Kigenere with shortrepeating LeFword. the old oAEeHtions to its Dse, whiHh Aoiled
down
tothe iZpossiAilitF of HorreHting a garAled dispatHh BDiHLlF enoDgh,Kanished with the telegraph. it
fDlfilled the reBDireZents of nonHoZ –proZisaAilitF of th e general sFsteZ and of ease of LeF Hhanges.
```

Step 7 — CRYPTOGRAPHY, MORE and USE

HIFUWNJITUQF (4 times) reads -R-PTOGRAPH-, which in a text about secret writing can only be CRYPTOGRAPHY. ZNIV then reads -ORE and is MORE, and DPV reads -SE and is USE.

```
key: ..fu.yncrg...odshwlapeti.m

the first one was waiting in the wings. this was poly –alphaAeticsuAstitution, in the form of the
straight–alphaAet Kigenere with shortrepeating Leyword. the old oAEections to its use, which Aoiled
down
tothe impossiAility of correcting a garAled dispatch BuicLly enough,Kanished with the telegraph. it
fulfilled the reBuirements of noncom –promisaAility of th e general system and of ease of Ley changes.
```

Step 8 — the rare letters

Seven letters are left, and each is fixed by a single word in which it is the only remaining unknown: XZUNPPXAXSXWF = IMPOSSIBILITY, IVBDXIVZVGWP = REQUIREMENTS, ZTENI = MAJOR, VTPXSFPNSKVO = EASILY SOLVED, LVFRNIO = KEYWORD, PUVHXTSXMVO = SPECIALIZED and VYHVUW = EXCEPT.

```
key: bqfujyncrgvkzodshwlapetixm

the first one was waiting in the wings. this was poly –alphabeticsubstitution, in the form of the
straight–alphabet vigenere with shortrepeating keyword. the old objections to its use, which boiled
down
tothe impossibility of correcting a garbled dispatch quickly enough,vanished with the telegraph. it
fulfilled the requirements of noncom –promisability of th e general system and of ease of key changes.
```

All 26 letters are now accounted for and the message reads as plain English.

5. The key and the message

Putting the two alphabets side by side gives the key, in the form of table 2.5 of the assignment:

```
ciphertext   a b c d e f g h i j k l m n o p q r s t u v w x y z
plaintext    b q f u j y n c r g v k z o d s h w l a p e t i x m
```

Read in the other direction — what each plaintext letter was enciphered to — the key is:

```
plaintext    a b c d e f g h i j k l m n o p q r s t u v w x y z
ciphertext   t a h o v c j q x e l s z g n u b i p w d k r y f m
```

A check on the answer

The second line is not a random permutation. Writing each letter as its position in the alphabet, the encipherment is

$$c = (7x + 19) \bmod 26$$

an affine cipher: `a` (0) goes to 19 = `t`, `b` (1) goes to 26 mod 26 = 0 = `a`, `c` (2) goes to 33 mod 26 = 7 = `h`, and so on for all 26 letters.

Nothing in the attack relied on this — the letters were recovered one at a time from frequencies and words, which is what the task asked for. But it is a strong confirmation that the solution is correct: a key that was even slightly wrong would show no such structure, so the odds of stumbling onto a clean arithmetic rule by accident are negligible.

The recovered message

```
The first one was waiting in the wings. This was poly -alphabeticsubstitution, in the form of the
straight-alphabet Vigenere with shortrepeating keyword. The old objections to its use, which boiled
down
tothe impossibility of correcting a garbled dispatch quickly enough,vanished with the telegraph. It
fulfilled the requirements of noncom -promisability of th e general system and of ease of key changes.
Moreover,it had the reputation of being unbreakable—which, if its cryptogramswere not divided into
words, it largely was. The military adopted it atonce.Then, in 1863, a retired Prussian infantry major
discovere d thegeneral solution for the periodic polyalpha -betic substitution. At onestroke he
demolished the only impregnable structure in cryptography.Signal officers, compelled to provide secure
communications, huntedfrantically for new field ciphers. They found many good ideas in thewritings of
the dilettante cryptographers who had proposed ciphers forthe protection of private messages. Soon some
of these systems wereserving in the various armies of Europe and the Americas. More ideascame from army
officers who had studied cryptography in the courses insignal communication that the national military
academies, such as St.Cyr, had added in the mid-1800s. Inevitably, cryptanalysts—who wereeither
amateurs
or soldiers with a professional interest, for fullprofessionals there were none—replied with new
techniques for breakingthe new ciphers. From the slow crawl of nomenclator days, when theintroduction
of
a special group meaning Disregard the preceding groupwould constitute a remarkable technical advance,
the race betweenoffense and defense in cryptology acclerated to its modern pace.The history of
cryptology from the decade that saw both the death ofthe black chambers and the birth of the telegraph
to World War I is thusa story of internal development. Without Rossignol s or Willeses, andwithout
major
wars or diplomatic struggles, cryptology could notinfluence world events, and, except for one or two
unusual cases, it didnot. The telegraph launched this evolution of cryptology. It broke the monopoly of
the nomenclator. Th enomenclator had reigned for 450 years as a general, all-purpose system,but it
could
not meet the new requirements either of high -leveldiplomatic or military communications or of low
-level signalcommunications, which the telegraph had engendered. Each call ed for itsown kind of
cryptosystem, a specialized one. Signal officers ranked thesesystems in a hierarchy, rising from the
simple and flexible and easilysolved to the extensive and hard to solve. The telegraph thus
stimulatedthe invention of many new ciphers and, by reaction, many new methodsof crypt- analysis, and
compelled their arrangement in a scale ofcomplexity.Many of these ciphers and techniques have become
classic and are inuse today. Moreover, cryptography still functions through a hierarchyand employs a
multitude of special systems. The telegraph therebyfurnished cryptography with the structure and the
content that it stillhas. It made it what it is today. All these things have antecedents, and just as
the telegraph itself did,so were there precursors of the cryptographic systems that itengendered.One
cipher system invented before the telegraph was so far ahead ofits time, and so much in the spirit of
the later inventions, that it deservesto be classed with them. Indeed, it deserves the front rank among
them,for this system was beyond doubt the most remarkable of all. So wellconceived was it that today,
more than a century and a half of rapidtechnological progress after its invention, it remains in active
use.But then it was invented by a remarkable m an, a well -known writer,agriculturalist, bibliophile,
architect, diplomat, gadgeteer, and statesmannamed Thomas Jefferson. He called it his "wheel cypher,"
and it seemslikely that he invented it either during 1790 to 1793 or during 1797 to1800.
```

The words that run together (`thegeneral`, `thewritings`) and the stray spaces inside words (`th e`, `call ed`) are inherited from the cryptogram as it was given, not introduced by the decryption — every one of them is present in the ciphertext in the same place.

The passage is an extract on how the telegraph reshaped nineteenth-century cryptology, ending on Thomas Jefferson's wheel cypher, from David Kahn's history *The Codebreakers*.

6. Implementation

The analysis was done with a small Python 3 program, written for this laboratory work, which uses only the standard library.

File	Contents
`english.py`	English letter frequencies (table 2.2) and the language model
`frequency.py`	letter, bigram, trigram and word statistics
`substitution.py`	applying a full or partial key to a text
`walkthrough.py`	the eight deductions above, as data
`solver.py`	automatic key recovery by hill climbing
`main.py`	the command-line program

```
python3 main.py analyse      # the frequency tables
python3 main.py walkthrough  # the attack, one deduction at a time
python3 main.py solve        # recover the key automatically
python3 main.py decrypt      # the plaintext
```

Showing a half-broken message

```
def partial_decrypt(text, key):
    out = []
    for char in text:
        if not char.isalpha():
            out.append(char)
            continue
        lowered = char.lower()
        if lowered in key:
            out.append(key[lowered])
        else:
            out.append(lowered.upper())
    return "".join(out)
```

This is what produces the intermediate texts in section 4. A letter already recovered is written in lowercase, a letter still unknown is left as the uppercase ciphertext letter, and anything that is not a letter is passed through untouched — which is what keeps the word boundaries and punctuation visible while the message is only half broken.

Repetition patterns

```
def pattern(word):
    seen = {}
    out = []
    for letter in word:
        if letter not in seen:
            seen[letter] = ALPHABET[len(seen)].upper()
        out.append(seen[letter])
    return "".join(out)
```

A substitution cannot change the shape of a word, so `pattern("wqvw")` and `pattern("that")` are both `ABCA`. Comparing patterns is what narrows a cipher word down to a handful of candidates, and it is the reason `XZUNPPXAXSXWF` could only be `IMPOSSIBILITY`.

Recovering the key automatically

The manual attack was also checked by a program that knows nothing about the message. It starts from the key that matches the two frequency orders — the opening move of the manual attack — and then improves it by swapping pairs of letters:

```
def _climb(key, scorer, rng):
    """Swap plaintext letters until no single swap improves the score."""
    best = dict(key)
    best_score = scorer.score(best)
    letters = list(ALPHABET)
    improved = True
    while improved:
        improved = False
        pairs = [(a, b) for i, a in enumerate(letters) for b in letters[i + 1:]]
        rng.shuffle(pairs)
        for first, second in pairs:
            candidate = dict(best)
            candidate[first], candidate[second] = best[second], best[first]
            candidate_score = scorer.score(candidate)
            if candidate_score > best_score:
                best, best_score = candidate, candidate_score
                improved = True
    return best, best_score
```

The score decides what counts as progress, and it combines two signals because neither is enough alone:

```
for word in self.words:
    plain = word.translate(table)
    padded = "_" + plain + "_"
    for index in range(len(padded) - 1):
        bigram_total += self.bigrams.get(padded[index:index + 2], self.floor)
    if plain in self.dictionary:
        word_total += len(plain) ** 2
```

Letter-pair statistics give a usable gradient while the text is still gibberish, when no real words have appeared yet; counting words of the dictionary sharpens the last steps, weighted by the square of the word length because a long word matching is far stronger evidence than a short one. Hill climbing can still stall in a local maximum, so the search restarts from shuffled keys and keeps the best result.

Run on this cryptogram the program returns `bqfujyncrgvkzodshwlapetixm` — the same key recovered by hand, and the main confirmation that the manual attack did not go wrong anywhere.

Checks

Seventeen automated checks cover the pieces: that the key is a one-to-one mapping of all 26 letters, that only letters are substituted and capitalisation is preserved, that every step of the walkthrough only adds letters and that the eight steps end at the recovered key, and that the plaintext contains the phrases it should. Two of them are worth singling out:

- the solver is given the recovered plaintext re-enciphered with a *random* key and has to break that too, which tests the method rather than the answer;
- the key is checked against the affine rule `c = (7x + 19) mod 26` for all 26 letters.

```
$ python3 -m unittest discover -s tests
.................
----------------------------------------------------------------------
Ran 17 tests in 7.632s

OK
```

7. Conclusions

The cipher has $26! \approx 4.03 \times 10^{26}$ possible keys and was broken in eight steps, by hand, without trying a single one of them. That is the whole lesson of the laboratory work: the size of the key space says almost nothing about the strength of a cipher. What matters is whether the ciphertext still carries the structure of the language, and a monoalphabetic substitution carries nearly all of it.

The attack was not really a search. Each step was a hypothesis that could be read back and checked immediately — if `WQV` had not been `THE`, the next line would have been nonsense and the guess abandoned at no cost. Because the evidence accumulates, the work gets easier as it goes: the first deduction needed the frequency table, and the last seven letters fell out of single words in which they were the only unknown.

Two details were worth noticing in practice. First, the frequency table is a way in, not an answer: lining up the two orders rank for rank was correct for only 9 of the 26 letters. It was correct for `E`, `T`, `A` and `O`, and that was enough to make the text readable; everything after that came from words, not counts. Second, what gave the message away fastest was not the letters at all but what was left *around* them — the word boundaries, the punctuation and the capitalisation. `WQV` standing alone 43 times is a much louder signal than any frequency, and a cipher that stripped the spaces would have been appreciably harder work.

The practical conclusion is that a monoalphabetic cipher is unusable for anything real. Hiding the word boundaries, padding the text or enciphering in fixed-length groups would slow an attacker down but would not fix the underlying flaw, since one plaintext letter still always produces the same ciphertext letter. The flaw is only repaired by making the substitution change as the message goes on — which is what polyalphabetic ciphers do, and which is the next thing to study.

8. Source code

The full source code is in the repository github.com/alo42/CS, in the `LAB2` directory.

```
git clone git@github.com:alo42/CS.git
cd CS/LAB2
python3 main.py walkthrough
```

`solve` additionally needs an English word list; it uses `/usr/share/dict/words`, present on macOS and most Linux systems, and the environment variable `ENGLISH_WORD_LIST` overrides the path. Everything else needs only Python 3 and the standard library.